Rabbits reproducing

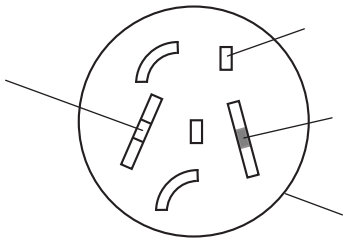
Rabbit genes

Imagine a group of rabbits. Each one has a gene for dark fur and a gene for light fur.

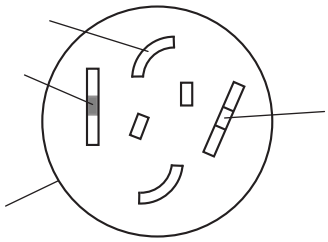
Word bank

- chromosome
- light fur gene
- dark fur gene
- light fur gene
- dark fur gene
- cell membrane
- Female body cell
- Male body cell

1 Simplified diagrams of body cells from a female rabbit and a male rabbit are shown below. Use the word bank to label the diagrams.



_____



_____

2 Refer to the diagrams of the body cells above to answer the following questions.

a How many chromosomes are in each body cell?_____

b Does each chromosome have a matching partner?_____

c How many pairs of chromosomes are there in each body cell?_____

d If all the rabbits have dark fur, which gene is dominant and which recessive?

_____

3 Rabbits actually have 44 chromosomes in each body cell, but only three pairs of chromosomes have been shown in the diagrams above for the sake of simplicity.

a During meiosis, gametes (sex cells) are produced. Male gametes are called _____ and female gametes are called _____.

b In the simplified version of rabbit body cells shown in the diagrams above, how many chromosomes would each gamete have?_____

c Is it possible for two male gametes to have different fur colour genes?_____

d Is it possible for two female gametes to have different fur colour genes?_____

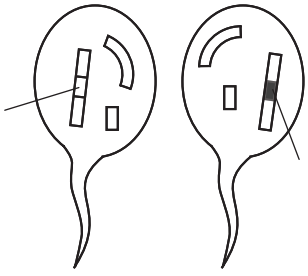
e Out of 100 eggs, how many would have a dark fur gene?_____

f Out of 100 eggs, how many would have a light fur gene?_____

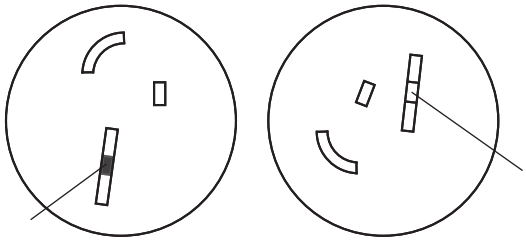
4 Simplified diagrams of sex cells from a female rabbit and a male rabbit are shown below. Use the word bank to label the diagrams.

Word bank

- light fur gene
- dark fur gene
- light fur gene
- dark fur gene
- cell membrane
- Types of sperm
- Types of eggs



_____



_____

5 Is there any way of predicting which type of sperm will fertilise each type of egg?_____

6 In your opinion, how many of 100 baby rabbits from the above parents would have:

a dark fur?_____ **b** light fur?_____

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